Refresher: How binary and categorical variables are encoded

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0

Refresher: How binary and categorical variables are encoded

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION
T-shirt/top	0
Trouser	1
Pullover	2
Dress	3
Coat	4
Sandal	5
Shirt	6
Sneaker	7
Bag	8
Ankle boot	9

Refresher: How binary and categorical variables are encoded

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION	ONE-HOT ENCODED VERSION									
T-shirt/top	0	1	0	0	0	0	0	0	0	0	0
Trouser	1	0	1	0	0	0	0	0	0	0	0
Pullover	2	0	0	1	0	0	0	0	0	0	0
Dress	3	0	0	0	1	0	0	0	0	0	0
Coat	4	0	0	0	0	1	0	0	0	0	0
Sandal	5	0	0	0	0	0	1	0	0	0	0
Shirt	6	0	0	0	0	0	0	1	0	0	0
Sneaker	7	0	0	0	0	0	0	0	1	0	0
Bag	8	0	0	0	0	0	0	0	0	1	0
Ankle boot	9	0	0	0	0	0	0	0	0	0	1

Important: Pick the Keras crossentropy loss function that matches the encoding

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0



binary_crossentropy

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION	ONE-HOT ENCODED VERSION									
T-shirt/top	0	1	0	0	0	0	0	0	0	0	0
Trouser	1	0	1	0	0	0	0	0	0	0	0
Pullover	2	0	0	1	0	0	0	0	0	0	0
Dress	3	0	0	0	1	0	0	0	0	0	0
Coat	4	0	0	0	0	1	0	0	0	0	0
Sandal	5	0	0	0	0	0	1	0	0	0	0
Shirt	6	0	0	0	0	0	0	1	0	0	0
Sneaker	7	0	0	0	0	0	0	0	1	0	0
Bag	8	0	0	0	0	0	0	0	0	1	0
Ankle boot	9	0	0	0	0	0	0	0	0	0	1

Important: Pick the Keras crossentropy loss function that matches the encoding

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0



binary_crossentropy

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION	ONE-HOT ENCODED VERSION									
T-shirt/top	0	1	0	0	0	0	0	0	0	0	0
Trouser	1	0	1	0	0	0	0	0	0	0	0
Pullover	2	0	0	1	0	0	0	0	0	0	0
Dress	3	0	0	0	1	0	0	0	0	0	0
Coat	4	0	0	0	0	1	0	0	0	0	0
Sandal	5	0	0	0	0	0	1	0	0	0	0
Shirt	6	0	0	0	0	0	0	1	0	0	0
Sneaker	7	0	0	0	0	0	0	0	1	0	0
Bag	8	0	0	0	0	0	0	0	0	1	0
Ankle boot	9	0	0	0	0	0	0	0	0	0	1



sparse_categorical_crossentropy

Important: Pick the Keras crossentropy loss function that matches the encoding

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0

→ binary_crossentropy

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION	ONE-HOT ENCODED VERSION
T-shirt/top	0	1 0 0 0 0 0 0 0 0 0
Trouser	1	0 1 0 0 0 0 0 0 0 0
Pullover	2	0 0 1 0 0 0 0 0 0 0
Dress	3	0 0 0 1 0 0 0 0 0 0
Coat	4	0 0 0 0 1 0 0 0 0 0
Sandal	5	0 0 0 0 0 1 0 0 0 0
Shirt	6	0 0 0 0 0 0 1 0 0 0
Sneaker	7	0 0 0 0 0 0 0 1 0 0
Bag	8	0 0 0 0 0 0 0 0 1 0
Ankle boot	9	0 0 0 0 0 0 0 0 0 1

↓ sparse_categorical_crossentropy

↓ categorical_crossentropy

Important: Pick the Keras crossentropy loss function that matches the encoding

BINARY CLASSIFICATION EXAMPLE

RAW DATA	ONE-HOT ENCODED VERSION
Yes	1
No	0

→ binary_crossentropy

MULTI-CLASS CLASSIFICATION EXAMPLE

RAW DATA	SPARSE ENCODED VERSION	ONE-HOT ENCODED VERSION
T-shirt/top	0	1 0 0 0 0 0 0 0 0 0
Trouser	1	0 1 0 0 0 0 0 0 0 0
Pullover	2	0 0 1 0 0 0 0 0 0 0
Dress	3	0 0 0 1 0 0 0 0 0 0
Coat	4	0 0 0 0 1 0 0 0 0 0
Sandal	5	0 0 0 0 0 1 0 0 0 0
Shirt	6	0 0 0 0 0 0 1 0 0 0
Sneaker	7	0 0 0 0 0 0 0 1 0 0
Bag	8	0 0 0 0 0 0 0 0 1 0
Ankle boot	9	0 0 0 0 0 0 0 0 0 1

↓ sparse_categorical_crossentropy

↓ categorical_crossentropy

Summary: Loss functions for different output layers

Output Variable	Output Layer	Loss Function
Single number (regression with a single output)		Mean squared error
Single probability (binary classification)		Binary cross-entropy
Vector of *n* numbers (regression with multiple outputs)	Stack of 	Mean squared error
Vector of n probabilities that add up to 1 (multi-class classification)	Softmax	Categorical cross-entropy